

Java Programming

Why another programming language? Don't we have too many already?

Why Java?

What are the differences or advantages of Java?

A Short History of Java

1991: Sun Microsystems engineers, led by Patrick Naughton and James Gosling, a language for consumer devices, the Green project.

C++, Oak, Java

1992: “*7”, an intelligent remote control. Then a cable TV box, Green became “First Person, Inc.” Dissolved 1994.

1994: World wide web, Mosaic, Marc Andressen, University of Illinois; Netscape. HotJava running inside web browsers.

1996: Java 1.0

1997: 1.1 Inner classes

1998: 1.2 strictfp

2000: 1.3

2004: 5.0 Generic classes, “for each” loop, varargs,

autoboxing, metadata, enumerations,
static import

2006: 6

2011: 7

Switch with strings, diamond operator,
binary literals, exception handling,
enhancements

2014: 8

Lambda expressions, interfaces with default
methods, stream and date/time libraries

2017: 9

Modules, miscellaneous language and library
enhancements

2018: 11

Local variable type inference (`var`), HTTP
client, removal of Java FX, JNLP, Java EE
overlap, and CORBA

2021: 17

Today!

Switch expressions, text blocks, instanceof
pattern matching, records, sealed classes

Characteristics of the Java Programming Language

Simple

Object-Oriented

Architecture-Neutral

Portable

Interpreted

Multithreaded

Secure

Robust

High-Performance

Distributed

Dynamic

Common Misconceptions about Java

Is an extension of HTML

XML can do the same

Is easy to learn

Will become a universal language

Is just another programming language

Is proprietary

Is slow because interpreted

Program runs inside web page

Has security risk

More complex than JavaScript

Able to replace desktop with “Internet appliance”

Books

Required:

Core Java, Volume 1: Fundamentals, 12th edition.
Horstmann, Cay. Oracle Press, December 2021. ISBN:
978-0137673629.

Recommended:

-*Java Cookbook, Problems and Solutions for Java Developers*, fourth edition, Ian F. Darwin, O'Reilly, 2020,
ISBN: 978-1492072584

-*Java in a Nutshell*, a Desktop Quick Reference, seventh
edition, Benjamin J. Evans and David Flanagan, O'Reilly,
2019, ISBN 978-149203725

-*Data Structures & Algorithm Analysis in Java*, third
edition, Clifford A. Shaffer, Dover Publications, 2011, ISBN
978-0486485812

The Java Programming Environment

Installing the Java Development Kit

Downloading the JDK, Standard Edition:

oracle.com/javadownload

OpenJDK

jdk.java.net

Installing the software:

Mac: /Library/Java/JavaVirtualMachine/
jdk-17.0.x.jdk/Contents/Home

Windows: E.g., c:\Java\jdk-17.0.x

From now on, we'll refer to our installation directory
as *jdk*

Testing the installation:

On a terminal window:

`javac --version`

Output:

`javac 17.0.4`

Additional setting for Windows: Book, p. 19

Installing Source Files and Documentation

-Source Files:

Unpacking compressed file `jdk/lib/src.zip`:

Make a directory in home:

`mkdir javasrc`

Locate `src.zip` and unzip it into `javasrc` directory:

`cd javasrc`

`jar xvf jdk/lib/src.zip`

-Documentation:

Download from

www.oracle.com/technetwork/java/javase/downloads

The download documentation file is

`jdk-17.0.x_doc-all.zip`

Unzip it:

`jar xvf Downloads/jdk-17.0.x_doc-all.zip`

Using the Command-Line Tools

-To compile the Helloworld class in Helloworld.java:

`javac Helloworld.java`

-To run the compile Helloworld Java class:

`java Helloworld`

Using an Integrated Development Environment

Eclipse, IntelliJ IDEA, NetBeans

Download Eclipse: eclipse.org/downloads

Writing a program in Eclipse:

- File, New, Project, Java Project

- Navigate to the project Location, Finish

- On left pane, Double-click on <filename>.java

- Click on project name, Run, Run-As, Java Application

- Program output is on bottom consol pane

- Errors, if any, are shown on error pane below source code

Fundamental Programming Structures

Comments

```
/* */  
//
```

Data Types

integer, floating-point, char, boolean

Data Type: Integer

byte	1 byte	-128 to 127
short	2	-32768 to 32767
int	4	-2,147,483,648 to 2,147,483,647
long	8	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807

Data Type: Floating Point

float 4 $\pm 3.40282347 \times 10^{38}$

double 8 $\pm 1.79769313486231750 \times 10^{308}$

Suffix for float: F or f, e.g., 3.14F

Without suffix, default is double

Suffix for exponent: E, e.g., 1.483E4 is 14830

In hexadecimal, using p for exponent, because
“E” is a hexadecimal digit

Special floating-point values:

Positive infinity

Double.POSITIVE_INFINITY

E.g., number/o

Negative infinity

Double.NEGATIVE_INFINITY

NaN, Not a Number

Double.NaN

E.g., 0/0, square root of negative

Caution of round-off errors. E.g., $2.0 - 1.1$ gives 0.89999...